

Web Application Security Testing using Kali Linux

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Role:- Cybersecurity Analyst Intern

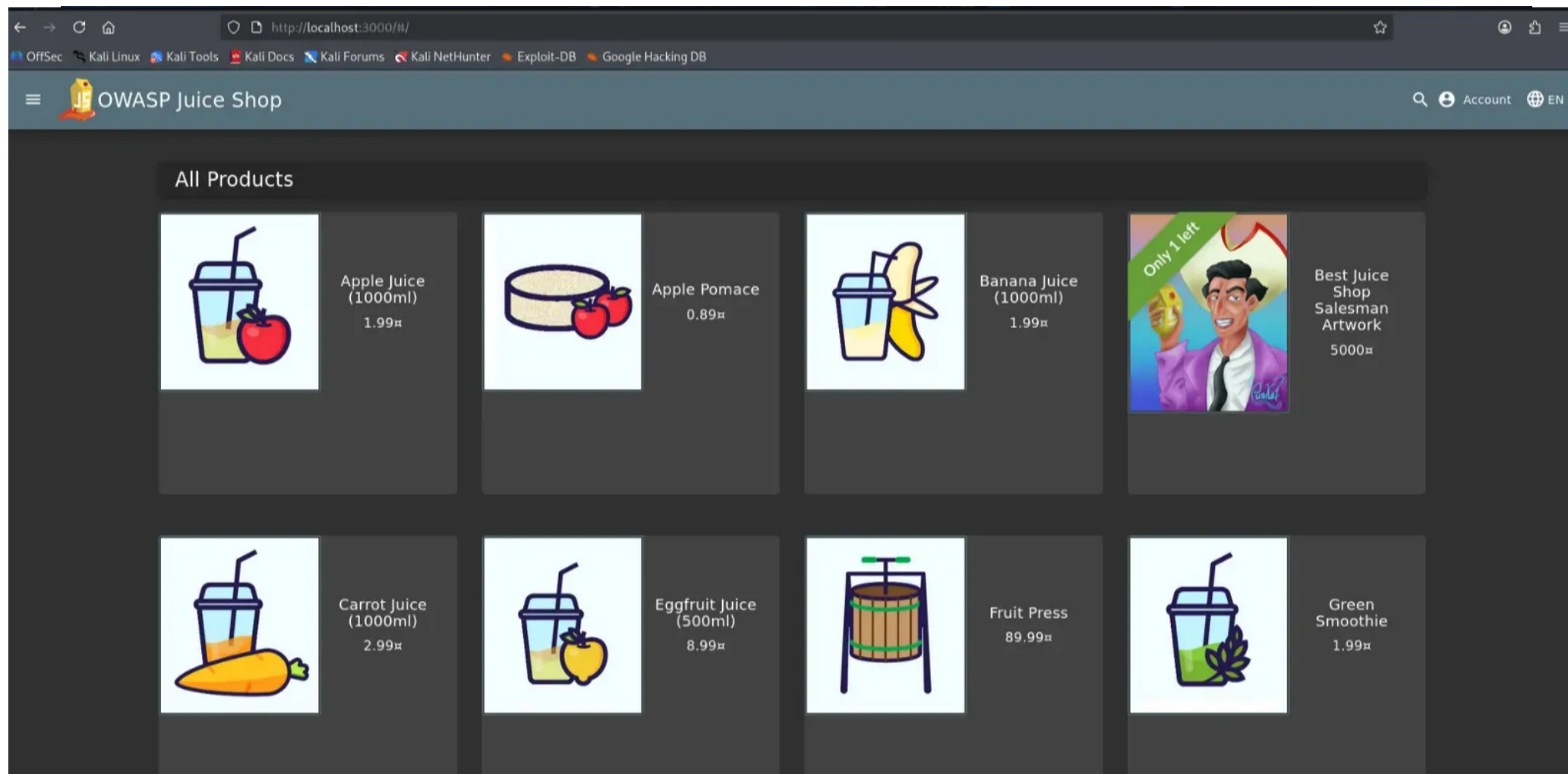
Date:- 29-04-2026

Objective:

- To To understand basic web security testing
- To analyze network and application traffic
- To identify vulnerabilities

Tools	Details
Attacker OS	Kali Linux (VMware)
Target App	OWASP Juice Shop
RAM Allocated	4 GB
CPUs	4
Network	NAT(Internet) + Host-Only(Lab)

●Juice-Shop Homepage



● Docker PS(Juice-Shop Running)

```
(sabhya@kali)-[~]  
$ sudo docker ps  
[sudo] password for sabhya:  
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                                                                 NAMES  
cbb638bac69f   bkimminich/juice-shop   "/nodejs/bin/node /j..."   3 hours ago   Up 3 hours   0.0.0.0:3000→3000/tcp, [::]:3000→3000/tcp   juice
```

● Nmap Scan(Port 3000 Open)

```
(sabhya@kali)-[~]  
$ nmap -p 3000 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-28 22:19 +0530  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.00025s latency).  
  
PORT      STATE SERVICE  
3000/tcp  open  ppp  
  
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
```

●IP Configuration(Kali Linux)

```
(sabhya@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:f3:23:7a brd ff:ff:ff:ff:ff:ff  
    inet 10.11.10.204/20 brd 10.11.15.255 scope global dynamic noprefixroute eth0  
        valid_lft 953sec preferred_lft 953sec  
    inet6 fe80::a00:27ff:fef3:237a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
5: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default  
    link/ether f2:81:ea:ae:a0:f9 brd ff:ff:ff:ff:ff:ff  
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::f081:eaff:feae:a0f9/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
6: vethaba1237@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default  
    link/ether a6:3d:69:fc:c5:a8 brd ff:ff:ff:ff:ff:ff link-netnsid 0  
    inet6 fe80::a43d:69ff:fe8c:c5a8/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever
```

●Burp Suite(HTTP History)

Intercept HTTP history WebSockets history Match and replace Proxy settings																		
Filter settings: Hiding CSS and image content; hiding specific extensions															Pro version only		Filter on	
#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start response ti	
1	http://localhost:3000	GET	/			200	75471	HTML		OWASP Juice Shop			127.0.0.1		19:30:34 28 Apr 2...	8080	8	
11	http://localhost:3000	GET	/assets/18n/en.json			200	33643	JSON	json				127.0.0.1		19:32:59 28 Apr 20...	8080	85	
12	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPd...	✓		200	326	text	io/				127.0.0.1		19:32:59 28 Apr 20...	8080	11	
13	http://localhost:3000	GET	/rest/admin/application-configuration			200	22159	JSON					127.0.0.1		19:32:59 28 Apr 20...	8080	81	
14	http://localhost:3000	GET	/rest/admin/application-version			200	404	JSON					127.0.0.1		19:32:59 28 Apr 20...	8080	20	
15	http://localhost:3000	GET	/api/Challenges/?name=Score%20Board	✓		200	835	JSON					127.0.0.1		19:32:59 28 Apr 20...	8080	603	
16	http://localhost:3000	GET	/rest/languages			200	4623	JSON					127.0.0.1		19:32:59 28 Apr 20...	8080	215	
18	http://localhost:3000	GET	/api/Quantities/			200	6646	JSON					127.0.0.1		19:33:32 28 Apr 20...	8080	635	
20	http://localhost:3000	GET	/rest/products/search?q=	✓		200	14033	JSON					127.0.0.1		19:33:32 28 Apr 20...	8080	334	
21	http://localhost:3000	GET	/rest/admin/application-configuration			200	22159	JSON					127.0.0.1		19:33:32 28 Apr 20...	8080	69	
22	http://localhost:3000	GET	/api/Challenges/?name=Score%20Board	✓		200	835	JSON					127.0.0.1		19:33:32 28 Apr 20...	8080	536	
23	http://localhost:3000	GET	/rest/admin/application-configuration			200	22159	JSON					127.0.0.1		19:33:33 28 Apr 20...	8080	32	
24	http://localhost:3000	GET	/rest/admin/application-configuration			200	22159	JSON					127.0.0.1		19:33:33 28 Apr 20...	8080	9	
25	http://localhost:3000	GET	/rest/admin/application-configuration			200	22159	JSON					127.0.0.1		19:33:33 28 Apr 20...	8080	18	
26	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPLi...	✓		200	326	text	io/				127.0.0.1		19:33:33 28 Apr 20...	8080	2	
27	http://localhost:3000	GET	/rest/admin/application-version			200	404	JSON					127.0.0.1		19:33:33 28 Apr 20...	8080	4	
28	http://localhost:3000	GET	/rest/admin/application-configuration			304	306						127.0.0.1		19:33:33 28 Apr 20...	8080	7	
30	http://localhost:3000	POST	/socket.io/?EIO=4&transport=polling&t=PKPL...	✓		200	215	HTML	io/				127.0.0.1		19:33:33 28 Apr 20...	8080	28	
31	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPL...	✓		200	262	text	io/				127.0.0.1		19:33:33 28 Apr 20...	8080	4	
32	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKP...	✓		200	230	text	io/				127.0.0.1		19:33:33 28 Apr 20...	8080	271	
33	http://localhost:3000	GET	/socket.io/?EIO=4&transport=websocket&sid=...	✓		101	129		io/				127.0.0.1		19:33:33 28 Apr 20...	8080	2	
46	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPLt...	✓					io/				127.0.0.1		19:34:04 28 Apr 2...	8080		
47	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPz...	✓					io/				127.0.0.1		19:34:27 28 Apr 20...	8080		
48	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPQ...	✓					io/				127.0.0.1		19:34:51 28 Apr 20...	8080		
49	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPQ...	✓					io/				127.0.0.1		19:35:16 28 Apr 20...	8080		
50	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPQ...	✓					io/				127.0.0.1		19:35:41 28 Apr 20...	8080		
51	http://localhost:3000	GET	/socket.io/?EIO=4&transport=polling&t=PKPQ...	✓					io/				127.0.0.1		19:36:06 28 Apr 20...	8080		

●Wireshark(HTTP Capture)

Capturing from Loopback: lo

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
28	24.583750308	127.0.0.1	127.0.0.1	HTTP	407	GET / HTTP/1.1
33	24.587929239	127.0.0.1	127.0.0.1	HTTP	407	GET / HTTP/1.1
37	24.932393416	127.0.0.1	127.0.0.1	HTTP	282	HTTP/1.1 200 OK
42	24.943719192	127.0.0.1	127.0.0.1	HTTP	282	HTTP/1.1 200 OK
73	40.818485700	127.0.0.1	127.0.0.1	HTTP	613	GET / HTTP/1.1
75	40.822943723	127.0.0.1	127.0.0.1	HTTP	459	HTTP/1.1 200 OK
77	41.193504562	127.0.0.1	127.0.0.1	HTTP	549	GET / HTTP/1.1
81	41.199690011	127.0.0.1	127.0.0.1	HTTP	550	GET / HTTP/1.1
83	41.200311838	127.0.0.1	127.0.0.1	HTTP	457	HTTP/1.1 200 OK
94	41.205302160	127.0.0.1	127.0.0.1	HTTP	549	GET / HTTP/1.1
96	41.205421245	127.0.0.1	127.0.0.1	HTTP	551	GET / HTTP/1.1

Frame 28: Packet, 407 bytes on wire (32000000) (capture length 407)

- Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 01:89:2b:6b:40:00 (01:89:2b:6b:40:00)
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Src Port: 54980, Dst Port: 80
- Hypertext Transfer Protocol

Hypertext Transfer Protocol: Protocol Packets: 628 - Displayed: 104 (16.6%) Profile: Default

●IP Configuration

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    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
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2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:f3:23:7a brd ff:ff:ff:ff:ff:ff  
    inet 10.11.10.204/20 brd 10.11.15.255 scope global dynamic noprefixroute eth0  
        valid_lft 953sec preferred_lft 953sec  
    inet6 fe80::a00:27ff:fef3:237a/64 scope link noprefixroute  
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    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
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6: vethaba1237@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default  
    link/ether a6:3d:69:fc:c5:a8 brd ff:ff:ff:ff:ff:ff link-netnsid 0  
    inet6 fe80::a43d:69ff:fefc:c5a8/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever
```


Command Used

Purpose	Command
Update System	<code>sudo apt update && sudo apt upgrade -y</code>
Install Docker	<code>sudo apt install docker.io -y</code>
Run Juice Shop	<code>sudo docker run -d -p 3000:3000 --name juice-shop bkimminich/juice-shop</code>
Check Docker status	<code>sudo docker ps</code>
Nmap Scan	<code>Nmap -p 3000 127.0.0.1</code>
IP Config	<code>ip a</code>
Burp Suite	<code>burpsuite</code>
Wireshark	<code>wireshark</code>

Conclusion

- Successfully built a personal cybersecurity lab with:
 - ✓ Kali Linux attacker machine
 - ✓ Juice Shop vulnerable target
 - ✓ Connectivity verified via Nmap
 - ✓ Burp Suite intercept working
 - ✓ Wireshark HTTP capture working